

To Grad or Not to Grad?
An Econometric Approach to Estimating the Return
to Graduate Degrees Relative to Bachelor's Degrees
Across Canada

By

Avery Bennett

University of Waterloo
ECON 421: Econometric Theory
Professor: Tao Chen

April 23, 2026

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1 Introduction

1.1 Motivations

The decision to pursue graduate studies is perhaps the most significant financial and personal gamble a student will face in their early career. It’s a dilemma familiar to any student who has sat in an upper year lecture, surrounded by ambitious peers deliberating their futures. Some will immediately head to law school or an MBA program, while others dive straight into the workforce. A few brave ones commit to five years of doctoral study. But who made the right choice?

In 2020, roughly 226,971 Canadians earned a bachelor’s degree and contemplated whether to pursue graduate education (Statistics Canada, 2025). Students in this position instinctively weigh the trade-offs, but in the absence of a definite payoff, it becomes rather difficult to determine what is truly “worth it.” For instance, Graduate programs require substantial investments of time, money, and an opportunity cost of delayed career entry, however the return remains uncertain. Does a master’s degree justify these costs? Will PhD holders earn enough to compensate for their extended time out of the labour market? And do these answers vary depending on whether you plan to work in Ontario, Nova Scotia, or Alberta?

1.2 Contribution

Using microdata from Statistics Canada’s 2021 Census, this paper will estimate how returns to graduate education compare with a bachelor’s, with a particular focus on whether these returns are invariant across Canada, or a dynamic response to provincial labour market equilibria. These findings contribute to the Canadian returns to education literature by providing a provincial comparison of graduate degree returns which is underrepresented in recent research. From a policy perspective, my results inform prospective graduate stu-

dents enrolment decisions and suggest that provincial labour markets value credentials quite differently.

2 Literature Review

The foundation for analyzing returns to education lies in human capital theory. Developed by Gary Becker and Theodore Schultz, this theory recognizes education as an investment that improves worker productivity, suggesting that individuals attain higher education and incur the affiliated costs, with the expectation that future earnings will justify the investment (Eide & Showalter, 2010). Graduate education represents an application of this theory, as it requires opportunity and direct costs, with individuals choosing to enrol based on the expectation, subject to uncertainty, that the future benefits are valued higher than these costs.

To quantify this return, the Mincer earnings function has become the standard framework to estimate wage returns (Heckman et al., 2003). Jacob Mincer’s model relates log earnings to schooling and experience. He originally used 1960 U.S. Census data to find that each additional year of schooling raised earnings by approximately 10%. The model assumes linearity, in that every extra year of education raises earnings by the same percentage. (Clark & Nielsen, 2024). While seemingly restrictive, a 2014 study reported that the basic model is quite stable, validating its continued use as a benchmark (Montenegro & Patrinos, 2014).

Existing literature widely regards ability bias as a central identification challenge. OLS estimates tend to be biased upward, since high ability positively influences education and wage. This endogeneity problem has motivated instrumental variable approaches. For instance, David Card’s work used college proximity as instruments, and demonstrated that men who grew up near colleges with 4 year programs obtained more education and earned higher wages. His results showed that instead of correcting upward OLS bias, IV estimates were even higher (Card, 1993). Card cautioned that IV estimates identify average effects for

the group of individuals whose schooling decisions were influenced by the instrument, rather than identifying an average for the broader population. He also noted that estimates tend to be imprecise, as no instrument achieves true randomization (Card, 2001).

In the Canadian context, Canada’s labour market is defined by exceptionally high post-secondary attainment, largely due to the many college and trades options (Boothby & Drewes, 2006). Despite these diverse pathways, a persistent “skill premium” exists for university graduates. Census data has demonstrated that the wage differential for bachelor’s degrees increased steadily from 1980 to 2005 (Boudarbat, et al., 2010). However, these measured monetary returns likely represent a lower bound of the “true” return to schooling, as standard earnings functions typically omit non-monetary private benefits and social externalities (Cohn, 1997). That being said, Canadian educational investments have demonstrated modest to high returns once both market and non-market effects have been considered (Cohn, 1997).

3 Data

3.1 Data Source

This study uses the 2021 Canada Census of Population Public Use Microdata Individuals File, published by Statistics Canada, containing demographic, geographic, income, social, and education variables. Since this study specifically looks into the returns of higher education, I subset the sample to only include observations for completed bachelor’s and graduate degrees, excluding those in the trades, college, or non academic post-secondary pathways. This ensures a more comparable set to draw conclusions from, as these degrees follow a similar academic trajectory. For this reason, I also omit the three territories, due to different labour and education patterns. Non-available or non-applicable categories were assigned NA values. This leaves us with 127,765 observations.

3.2 Variable Selection

- **Dependent Variable:** For our dependent variable, we use the logarithm of wage. This is standard in Mincer equations for two primary reasons. Firstly, it allows for the coefficients to be interpreted as percentage changes, which is more economically meaningful when comparing wage returns. Secondly, the log transformation normalizes the distribution to address right-skewed wage data (Feng et al., 2014). The variable, *Wages*, refers to the gross wages and salaries before deductions, such as income taxes or pension plan contributions.
- **Independent Variables:** Our first independent variable of interest is *Graduate*, a binary variable taking a value of 1 if the individual has completed a graduate degree (includes Master’s and Doctoral degrees, as well as professional degrees in Law and Medicine), and 0 if their highest education is a completed bachelor’s degree. We are also interested in *Province*, a fixed effect variable, which includes 10 binary variables corresponding to each province. We set Ontario as the base.
- **Controls:** The remaining controls include *Age* (numerical, proxy for experience), *STEM* (science, technology, engineering, and math fields = 1, otherwise = 0), *Female* (female = 1, male = 0), *Immigrant* (immigrant = 1, Canadian born = 0), *NonPermRes* (non-permanent resident = 1, permanent resident = 0), *Indigenous* (Indigenous = 1, otherwise = 0), *LargeCity* (living in Toronto, Ottawa, Montreal, Calgary, Edmonton, or Vancouver = 1, otherwise = 0), and *FullTime* (employed in a full-time position = 1, otherwise = 0).

4 Methodology

The empirical method is motivated by Mincer specification, as discussed in the Literature Review. The model in this paper will build on the original by adding controls, province

fixed effects, and consider potential instruments to address endogeneity. Lastly, we will estimate a Tobit model due to censoring reported in the wage data. The results will be compared with the OLS estimates, to determine if the censoring has a significant effect on the estimators.

4.1 Ordinary Least Squares

Our primary model is defined as the following:

$$\log(wage_j) = \beta_0 + \beta_1 Graduate_j + \sum_{p=1}^P \delta_p Province_{jp} + \sum_{p=1}^P \gamma_p (Graduate_j \times Province_{jp}) + X_j' \beta + u_j$$

where $Graduate_j$ is a binary variable for the attainment of a graduate degree, $Province_{jp}$ represents province fixed effects, X_j is a vector of controls, and γ_p captures the interaction of graduate degree returns across provinces.

This model can be simplified to:

$$Y_j = X_j' \beta + u_j$$

The derivation of the OLS estimator $\hat{\beta}$ is:

$$\hat{\beta} = \arg \min_{\beta} \sum_{j=1}^n (Y_j - X_j' \beta)^2$$

Taking the first order condition and solving, we arrive at:

$$\hat{\beta} = \left(\sum_{j=1}^n X_j X_j' \right)^{-1} \left(\sum_{j=1}^n X_j Y_j \right)$$

Biasedness

To determine if the OLS estimator is biased, we take the expectation of $\hat{\beta}$:

$$E[\hat{\beta}] = E \left[\left(\frac{1}{n} \sum_{j=1}^n X_j X_j' \right)^{-1} \left(\frac{1}{n} \sum_{j=1}^n X_j Y_j \right) \right]$$
$$E[\hat{\beta}] = \beta + E \left[\left(\frac{1}{n} \sum_{j=1}^n X_j X_j' \right)^{-1} \left(\frac{1}{n} \sum_{j=1}^n X_j u_j \right) \right]$$

If X is endogenous, then $E[Xu] \neq 0$, meaning the estimation uncertainty is not zero and our estimator is biased. By the Law of Large Numbers:

$$\hat{\beta} \xrightarrow{p} \beta + (E[XX'])^{-1} E[Xu]$$

Asymptotic Normality

One of the assumptions for OLS is homoskedasticity. Thus, a Breusch-Pagan test was run to determine how to construct appropriate standard errors. We strongly reject homoskedasticity ($p < 0.00$), and report heteroskedasticity-robust standard errors.

By the Law of Large Numbers and Central Limit Theorem, the asymptotic normality is:

$$\sqrt{n}(\hat{\beta} - \beta) = \left(\frac{1}{n} \sum_{j=1}^n X_j X_j' \right)^{-1} \left(\frac{1}{\sqrt{n}} \sum_{j=1}^n X_j u_j \right) \xrightarrow{d} \mathcal{N} \left(0, (E[XX'])^{-1} E[XX'u^2] (E[XX'])^{-1} \right)$$

4.2 Tobit Model Approach

The Census documentation noted that the variable *Wage* had instances of top coding and bottom coding. This distorts both ends of the distribution by treating them as having identical income values regardless of their true earnings. Specifically, the lower end is bottom coded at \$1 (1.7%) and the upper end, top coded at \$967,998 (0.3%). One way to address

censoring is by Tobit estimation, an MLE method that treats the observed $\log(\text{wage})$ as censored realizations of a latent variable Y_i^* . Although the proportion of censored observations is relatively small, the Tobit model is employed as a robustness check to ensure that the OLS estimates are not significantly biased by the restricted range of the dependent variable. Under the assumptions of asymptotic normality, homoskedasticity, and i.i.d. errors, it provides consistent estimates of the independent variables on latent wages. The following figure is a histogram of the wage distribution.

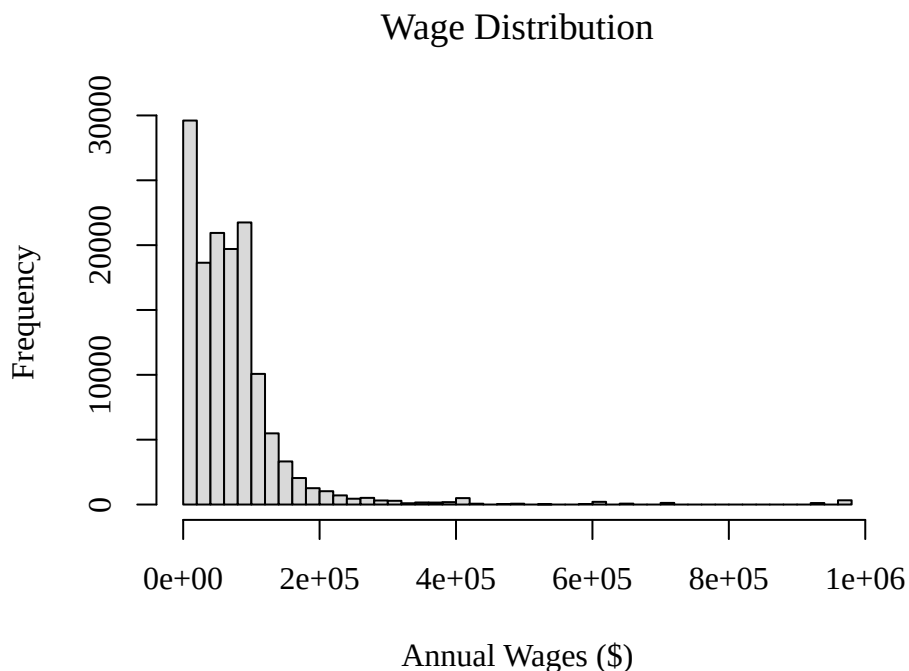


Figure 1: Distribution of Annual Wages

The distribution is right-skewed and exhibits censoring at both extremes. As shown in Figure 1, the impact of censoring is visually more prominent at the lower bound. Scientific notation on the x-axis represents wages in Canadian dollars (e.g., 2e+05 denotes \$200,000).

Tobit Model Derivation

The latent wage, Y_j^* , represents the true earnings of an individual but is only observed within the range $[B_L, B_U]$. The latent variable follows the model:

$$Y_j^* = X_j' \beta + u_j, \quad u_j \sim \mathcal{N}(0, \sigma^2)$$

where,

$$Y_j = \begin{cases} B_L & \text{if } Y_j^* \leq B_L \\ Y_j^* & \text{if } B_L < Y_j^* < B_U \\ B_U & \text{if } Y_j^* \geq B_U \end{cases}$$

The log-likelihood is:

$$\begin{aligned} \ell(\beta, \sigma) = \frac{1}{n} \sum_{j=1}^n & \left[\mathbf{1}(B_L < Y_j < B_U) \log \left(\frac{1}{\sigma} \phi \left(\frac{Y_j - X_j' \beta}{\sigma} \right) \right) \right. \\ & \left. + \mathbf{1}(Y_j = B_L) \log \left(\Phi \left(\frac{B_L - X_j' \beta}{\sigma} \right) \right) + \mathbf{1}(Y_j = B_U) \log \left(1 - \Phi \left(\frac{B_U - X_j' \beta}{\sigma} \right) \right) \right] \end{aligned}$$

Maximizing the log-likelihood with respect to β and σ yields the Tobit estimators.

4.3 Addressing the Endogeneity of Graduate

It can be argued quite easily that our education measurement, *Graduate*, is endogenous. One argument is that it is a choice variable. That is, individuals make their own decision on how much, and what type of, education to pursue, and often base this choice on unobserved influences. Many of these influences, notably ability, access, preferences, and family influence are captured in the error term, creating omitted variable bias. Additionally, individuals may be incentivized to acquire higher education, with the anticipation of higher future wages, creating simultaneity. Measurement errors present another source of endogeneity. Since

census data is self-reported, there is a potential for misclassification. By way of illustration, graduate students who are working but have not yet earned their degree may be incorrectly categorized. This endogeneity prevents us from observing a causal relationship, as coefficients are biased (likely upwards) and inconsistent. All OLS regression results will be interpreted as correlations.

Ideally, we would use two instruments to capture the part of *Graduate* that is not correlated with the error, however, two problems are encountered in this process. First, finding instruments that satisfy validity and relevance in census data is difficult. This is because most variables in the dataset that are correlated with education, affect wages directly. When we do find exogenous variables, they are often not correlated enough with education. Second, the interaction of *Graduate* and *Province* creates ten endogenous variables that must be instrumented for, instead of just one.

The two stage least squares estimation was attempted. An extensive search for suitable instruments within the 2021 Census PUMF was conducted, and the instruments selected were *Parental Place of Birth* (primary household parent born outside of Canada = 1, otherwise = 0) and *Second Generation* (second generation = 1, otherwise = 0). The Hausman test rejected the exogeneity of *Graduate* ($p=0.003$), and the Sargan test failed to reject the exogeneity of the instruments ($p=0.346$). The F-test reported the instruments were strong for *Graduate:ProvinceQC* ($F=54.723$), *Graduate:ProvinceAB* ($F=18.876$), *Graduate:ProvinceBC* ($F=16.949$), and *Graduate:ProvinceNS* ($F=10.100$), but weak for *Graduate* ($F=5.057$), *Graduate:ProvincePEI* ($F=1.503$), *Graduate:ProvinceNL* ($F=3.888$), *Graduate:ProvinceSK* ($F=6.585$), *Graduate:ProvinceMB* ($F=3.751$), and *Graduate:ProvinceNB* ($F=3.934$).

Despite the validity from the Sargan test, economic reasoning makes it possible that family background does not influence education enough. In fact, recent literature has cautioned the use of family background instruments as they are typically considered “weak or invalid”

(Clark & Nielsen, 2024). Given the variables to choose from in census data, parental place of birth and second generation status had the strongest results as instrument contenders, but not strong enough to implement 2SLS inference.

5 Results

All regression estimates in this section are interpreted as correlations. I will use the $(e^\beta - 1)$ transformation, report interactions as partial effects, and for all observations, hold the remaining regressors constant, where the vector of regressors is *Graduate*, *Province*, *STEM*, *Age*, *Female*, *LargeCity*, *FullTime*, *Immigrant*, *NonPermRes*, and *Indigenous*.

5.1 Model 1: Ordinary Least Squares

Table 1: OLS Estimates for Determinants of Wage

	Estimate	Robust SE	t value	p-value
(Intercept)	8.484	0.027	317.404	0.000
Graduate	0.155	0.008	19.879	0.000
STEM	0.092	0.008	11.800	0.000
LargeCity	0.057	0.008	7.134	0.000
ProvinceNL	-0.096	0.039	-2.439	0.015
ProvincePEI	-0.059	0.037	-1.587	0.113
ProvinceNS	-0.168	0.024	-6.959	0.000
ProvinceNB	-0.094	0.025	-3.852	0.000
ProvinceQC	-0.090	0.009	-10.085	0.000
ProvinceMB	-0.036	0.018	-1.970	0.049
ProvinceSK	-0.035	0.025	-1.384	0.166
ProvinceAB	0.048	0.011	4.406	0.000
ProvinceBC	-0.036	0.010	-3.430	0.001
Age	0.096	0.002	41.693	0.000
Female	-0.150	0.007	-21.541	0.000
Immigrant	-0.283	0.008	-36.989	0.000
NonPermRes	-0.590	0.019	-31.148	0.000
Indigenous	-0.034	0.024	-1.397	0.162
FullTime	1.650	0.015	113.393	0.000
I(Age^2)	-0.001	0.000	-32.950	0.000

Note: N = 127,765, R-squared = 0.2357, Adj. R-squared = 0.2356, F-statistic = 2,074 (p < 0.001)

Table 1 presents the OLS estimates of our basic model. The regression results indicate that when Canadians hold a graduate degree, they earn, on average, 16.8% higher wages

than those with a bachelor's ($p<0.00$). Individuals who studied in a STEM major, earn, on average, 9.6% higher wages, than those in non-STEM disciplines ($p<0.00$). Individuals who live in a large city earn, on average, 5.9% higher wages than those who don't ($p<0.00$). The results demonstrate variation in wages between provinces, with most showing a lower wage return than in Ontario. Individuals in Alberta however, earn approximately 4.9% higher wages than those in Ontario ($p<0.00$). The marginal effect of *Age* demonstrates a positive, diminishing relationship, which is consistent with the existing literature ($p<0.00$). Wages increase at a diminishing rate until 47.3 years of age, after which it declines. Among all university degree holders, *Female* wages are, on average, 13.9% lower than males ($p<0.00$). *Immigrant* and *Non-Permanent Resident* status, both have negative partial effects, earning, on average, 24.6% ($p<0.00$) and 44.6% ($p<0.00$) less than non-immigrants and permanent residents. Canadians with *Indigenous* status also exhibit a negative partial effect, however it is not significant. *Full Time* employment is highly correlated with wage ($p<0.00$).

5.2 Model 2: Ordinary Least Squares With Interaction

Table 2: OLS Estimates for Returns to Education

	Estimate	Robust SE	t value	p-value
(Intercept)	8.475	0.027	314.182	0.000
Graduate	0.190	0.011	16.612	0.000
STEM	0.092	0.008	11.814	0.000
LargeCity	0.057	0.008	7.105	0.000
ProvinceNL	-0.078	0.043	-1.798	0.072
ProvincePEI	-0.103	0.045	-2.302	0.021
ProvinceNS	-0.175	0.027	-6.475	0.000
ProvinceNB	-0.111	0.028	-3.950	0.000
ProvinceQC	-0.072	0.010	-6.961	0.000
ProvinceMB	-0.014	0.020	-0.681	0.496
ProvinceSK	-0.009	0.028	-0.308	0.758
ProvinceAB	0.078	0.012	6.459	0.000
ProvinceBC	-0.022	0.012	-1.867	0.062
Age	0.096	0.002	41.638	0.000
Female	-0.151	0.007	-21.557	0.000
Immigrant	-0.283	0.008	-37.025	0.000
NonPermRes	-0.589	0.019	-31.048	0.000
Indigenous	-0.035	0.024	-1.436	0.151
FullTime	1.650	0.015	113.421	0.000
I(Age^2)	-0.001	0.000	-32.927	0.000
Graduate:ProvinceNL	-0.061	0.092	-0.665	0.506
Graduate:ProvincePEI	0.173	0.076	2.280	0.023
Graduate:ProvinceNS	0.027	0.057	0.474	0.635
Graduate:ProvinceNB	0.082	0.055	1.509	0.131
Graduate:ProvinceQC	-0.061	0.020	-3.083	0.002
Graduate:ProvinceMB	-0.094	0.046	-2.056	0.040
Graduate:ProvinceSK	-0.108	0.064	-1.680	0.093
Graduate:ProvinceAB	-0.115	0.027	-4.335	0.000
Graduate:ProvinceBC	-0.047	0.024	-1.950	0.051

Note: N = 127,765, R-squared = 0.236, Adj. R-squared = 0.2358, F-statistic = 1,409 (p < 0.001)

Table 2 presents the OLS results of our primary model of focus. It differs from the previous as it includes the interaction between graduate and province, which is critical for observing heterogeneity between provinces. Based on the regression results, Canadians living in Ontario who hold a graduate degree, earn on average, 20.9% higher wages than those with a

bachelor's ($p < 0.00$). The results demonstrate heterogeneity between provinces, of individuals wages when they have a graduate degree. In the Maritime provinces, the average marginal effects are 13.8% ($p = 0.506$) in *Newfoundland & Labrador*, 43.8% ($p = 0.023$) in *Prince Edward Island*, 24.2% ($p = 0.635$) in *Nova Scotia*, and 31.3% ($p = 0.131$) in *New Brunswick*. Moving inward, these partial effects are 13.8% ($p = 0.002$) in *Quebec*, 10.1% ($p = 0.040$) in *Manitoba*, and 8.5% in *Saskatchewan* ($p = 0.093$). Finally, out west, we observe partial effects of 7.8% ($p < 0.00$) in *Alberta* and 15.4% ($p = 0.051$) in *British Columbia*.

5.3 Model 3: Tobit

Table 3: Tobit Estimates for Returns to Education

	Estimate	Robust SE	t value	p-value
(Intercept)	8.469	0.027	311.162	0.000
Graduate	0.192	0.012	16.585	0.000
STEM	0.091	0.008	11.655	0.000
LargeCity	0.059	0.008	7.317	0.000
ProvinceNL	-0.079	0.044	-1.812	0.070
ProvincePEI	-0.103	0.045	-2.311	0.021
ProvinceNS	-0.176	0.027	-6.473	0.000
ProvinceNB	-0.112	0.028	-3.959	0.000
ProvinceQC	-0.074	0.010	-7.108	0.000
ProvinceMB	-0.014	0.020	-0.675	0.500
ProvinceSK	-0.009	0.028	-0.338	0.736
ProvinceAB	0.076	0.012	6.283	0.000
ProvinceBC	-0.024	0.012	-1.994	0.046
Age	0.097	0.002	41.456	0.000
Female	-0.153	0.007	-21.690	0.000
Immigrant	-0.285	0.008	-36.919	0.000
NonPermRes	-0.590	0.019	-30.797	0.000
Indigenous	-0.035	0.024	-1.448	0.148
FullTime	1.654	0.015	112.415	0.000
I(Age^2)	-0.001	0.000	-32.814	0.000
Graduate:ProvinceNL	-0.065	0.093	-0.695	0.487
Graduate:ProvincePEI	0.170	0.075	2.266	0.023
Graduate:ProvinceNS	0.025	0.058	0.434	0.664
Graduate:ProvinceNB	0.080	0.055	1.470	0.141
Graduate:ProvinceQC	-0.064	0.020	-3.174	0.002
Graduate:ProvinceMB	-0.097	0.046	-2.103	0.036
Graduate:ProvinceSK	-0.110	0.064	-1.711	0.087
Graduate:ProvinceAB	-0.118	0.027	-4.413	0.000
Graduate:ProvinceBC	-0.049	0.024	-2.039	0.041
Log(scale)	0.178	0.008	22.036	0.000

Note: Log-Likelihood: -204,715.7, Scale Parameter (σ): 1.195, N: 127,765

Table 3 presents the MLE Tobit estimators. Based on this method, Canadians living in Ontario who hold a graduate degree, earn on average, 21.2% higher wages than those with a bachelor's ($p < 0.00$). In the Maritime provinces, the partial effects of graduate education on

wage is 13.5% ($p=0.487$) in *Newfoundland & Labrador*, 43.6% ($p=0.023$) in *Prince Edward Island*, 24.2% in *Nova Scotia* ($p=0.664$), and 31.3% ($p=0.141$) in *New Brunswick*. Moving inward, the partial effects are 13.7% ($p=0.002$) in *Quebec*, 10.0% ($p=0.036$) in *Manitoba*, and 8.5% in *Saskatchewan* ($p=0.087$). Finally, out west, we observe partial effects of 7.7% ($p<0.00$) in *Alberta* and 15.4% ($p=0.041$) in *British Columbia*. The convergence of the OLS and Tobit results despite the presence of top and bottom coding suggests that our estimates are not significantly influenced.

6 Discussion

6.1 Economic Interpretation of Marginal Effects

Our results demonstrate a pattern of positive wage returns to graduate degrees relative to bachelor’s degrees in Canada. The baseline OLS reported a pooled wage return that is about 16.8% higher than the return of a bachelor’s degree, and for STEM majors, an additional return about 9.6% higher than non-STEM majors. The return to higher education aligns with human capital theory, in that the labour market tends to reward the skills and innovation associated with advanced academic qualification.

The second model, which interacted *Graduate* with *Province*, is more related to the focus of this study, which is to estimate differences in graduate wages across geographic regions. In particular, the return associated with graduate degrees in Ontario (20.9%) was among the highest estimates nationally, which may be reflected by its labour market composition. Ontario is Canada’s most populated province, with over 90% of its population concentrated in the south. Widely regarded as Canada’s “economic hub,” southern Ontario is dominated by finance, insurance, consulting, law, and professional sectors (CanadaVisa, 2026). Oftentimes, these roles require a master’s degree, anywhere from an MBA, JD, or a specialized master’s, providing a plausible case for why the wage premium seems more systematic than in other

labour markets. This is consistent with economic signaling theory, which suggests that in the presence of asymmetric information, employers utilize graduate degrees as “signals” to identify capable candidates (Science Direct, n.d.).

Alberta’s estimates present the most striking evidence of heterogeneity in this sample. From the baseline, Alberta exhibits the highest wage return overall (8.0% higher than Ontario), yet is associated with the lowest marginal return to graduate education (7.8%). This paradox is consistent with several structural characteristics of the provincial economy. First, Alberta consistently reports the highest household median after-tax income in Canada (\$88,500 in 2023), as well as the highest real GDP per person (~\$72,000 in 2023) (Statistics Canada, 2023). Its renowned energy and natural resources industries are among the most productive sectors in Canada, suggesting that the marginal product of labour is higher in Alberta than in other provinces (Finlayson, 2026). This may also explain why skilled trades and field experience can outweigh academic qualification, keeping in mind that graduate degrees are still rewarded, but openings are not as common. A higher opportunity cost of pursuing graduate education is therefore very possible, when the alternative is so lucrative. This effect might extend to Alberta’s neighbouring Prairies, Manitoba and Saskatchewan, which rely heavily on agriculture and resource extraction. However, the resource sectors of these provinces don’t reach the same level of intensity as Alberta. This is reflected in a slightly higher marginal return to graduate education than in Alberta, although the lack of statistical significance in Saskatchewan necessitates some uncertainty.

Quebec’s graduate premium (13.7%) falls in between Ontario and Alberta, likely reflecting differences in labour market institutions. Quebec’s labour market is in many ways, unique within Canada, having the highest unionization rate (40% of all payroll employees in 2024), strong labour protections, and strict French language requirements enforced by Bill 101 (Note: this dataset was collected before Bill 96 was passed) (Institut de la statistique du Québec, 2025). The 1997 Canada-Quebec Labour Market Agreement in particular, transferred significant authority of employment management and training to the province.

In combination with the Civil Code of Quebec, these policies focused on pay equity within organizations, which may be associated with a less dispersed wage distribution, as well as job security and local priorities which may reduce accessibility for non-Francophone individuals. This structure is consistent with a reward for higher education, but with smaller wage gaps between education levels. The accessibility also aligns with our findings of an immigrant penalty (-24.6%), as Quebec’s system may pose higher barriers to labour market entry.

British Columbia’s graduate premium (15.4%) is much closer to Ontario than the Prairies. Its labour structure is often described as a “dual economy”, such that the northern is resource-based and the southern is technology-based (Caddell, 2018). According to Statistics Canada’s 2019 Labour Force Survey, occupations in construction, manufacturing, trade, professional services, educational services, and healthcare make up the majority of their labour force (Government of Canada, 2019). Thus, there exists a mix of sectors, some rewarding academics, and others, field experience where a bachelor’s degree is sufficient. This provides a plausible explanation of why the graduate premium remains comfortably higher than Alberta’s, but is unable to fully reach Ontario’s.

The Maritime provinces present similar results with one another. Prince Edward Island had an anomalously large estimated return of 43.8%, far exceeding Ontario. However, the standard error for the PEI interaction (0.076) is nearly seven times larger than Ontario (0.011). As Canada’s least populated province, the imprecision of this estimate likely reflects the limitations of a small sample size, in which graduate degree holders are concentrated among a handful of the province’s top earners. Nova Scotia (24.2%) and New Brunswick (31.3%) fell subject to a similar problem, with large yet statistically insignificant estimates. Newfoundland and Labrador seemed more sensible at 13.8%, but was insignificant. That said, New Brunswick and Nova Scotia have placed a focus on marine and renewable innovation while facing an aging population, which is theoretically consistent with a demand for graduate degrees and a positive wage correlation, however the estimates themselves are highly uncertain (SR&ED, 2025).

From an economic theory perspective, the variation in returns across provinces suggests that the wage premium for graduate education is largely dependent on the complementarity between the degree and the location. This idea is theoretically supported by Tarun Sabarwal’s (2023) framework of complementarities, which says that labour markets can settle into a lattice of multiple equilibria (Sabarwal, 2023). In this context, the lower returns in the Prairies are consistent with its environment, where degrees and experience act more as substitutes, whereas the high premiums in Ontario and B.C. can be interpreted as reflecting settings where advanced education and experience are strong complements. The variation also provides a compelling argument against pure ability bias, since even high ability individuals receive significantly different returns based on their province. Hence, the value of a graduate degree appears to be determined by the specific equilibrium of the province in which it is utilized.

6.2 Limitations and Future Improvements

Several limitations constrain the scope of these findings, with the most significant as the inability to claim causal identification. Since *Graduate* is a choice variable, the model is subject to selection bias, and therefore endogeneity. The available instrumental variables in this dataset were invalid, preventing the 2SLS method from producing something causal. It is possible that the OLS estimates overstate the causal effect as choice motivators like ability, among others, cannot be controlled for.

A second important limitation is that wages are observed at a single point in time, such that measurements of how graduate degree returns evolve over a given career, is not captured by cross-sectional data. Future research with panel data could trace the earnings pattern for degree holders, to determine how the estimates differ for longer horizons.

Moreover, this study uses 2021 Census data, which was taken in the 2020 reference period, a year that was irregularly affected by the COVID-19 pandemic. For many degree

holders, the pandemic generated shocks, such as unemployment, layoffs, emergency benefits (CERB), and other distortions which are not representative of “typical” labour market conditions.

6.3 Conclusion

This paper set out to answer a deceptively simple question. Does a graduate degree pay off in Canada, and does the answer depend on the province in which you reside? Our evidence suggests that it does, but unevenly.

The OLS and Tobit estimates consistently indicate a positive graduate wage return relative to bachelor’s degree holders. The more interesting story, however, is geographic. The interaction model reveals that the graduate premium is better described as a distribution that follows provincial labour markets, than as a fixed return. Ontario and the Maritime provinces tend to reward graduate credentials more generously than the Prairie provinces, particularly Alberta, which showed the smallest return.

The practical takeaway is that geographic location matters substantially for graduate degree returns. The premium exists, and is largest in the labour markets that complement graduate academic skills. Prospective students should consider both field of study and intended labour market when evaluating whether graduate education justifies its costs, recognizing that the approximate returns are correlations rather than guaranteed outcomes.

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